A Tiny Robot That Swims With Living Muscle

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MIT scientists have built a tiny swimming robot that moves with living muscle instead of a bulky motor. The robot is a thin, flexible gel frame about the length and width of a stick of gum, with two fins. Each fin is coated with muscle cells far thinner than a human hair, genetically modified to contract when light shines on them.

Shining light on one fin makes it flap and push the robot through water. By choosing which fin gets light, and how often, researchers can steer the robot and change its speed. In tests, it swam and turned through a simple underwater maze. At its fastest, it covered about four body lengths per minute, much slower than an Olympic swimmer's 65, but comparable to a cow shark swimming leisurely.

"It takes a lot of force to move through water versus air," said study author Ritu Raman, an associate professor of mechanical engineering at MIT. "The robot's quite strong, given its size."

The study, published in Advanced Functional Materials, describes the first very thin, two-dimensional robot that can move from place to place using living muscle tissue. Most existing **biohybrid** robots use larger, three-dimensional structures made from laboratory-grown **skeletal** muscle, usually requiring millions of cells. Raman said thinner designs could be cheaper to build and might move more efficiently. She believes muscle-powered robots could one day handle delicate jobs in places too fragile or unpredictable for ordinary hardware, because living tissue is soft, responsive, and can heal itself.

The swimmer builds on an earlier artificial muscle disk inspired by the human **iris**. That disk showed thin muscle layers could be grown in detailed patterns and controlled with light, but it moved only about 100 **microns**. Its **fibrin** gel, a very soft gel, could shrink under the muscle's force, so the team studied the gel's makeup, stiffness, and groove patterns to build a stronger foundation.

Word Treasure

biohybrid -- A machine that mixes living tissue with man-made parts.

skeletal -- Muscle of the kind that moves bones and the body.

fibrin -- A protein the team used to make a very soft gel.

microns -- A micron is a tiny unit of length, one millionth of a metre.

iris -- The coloured ring in the eye that controls how much light enters.

Background you need

A biohybrid robot is a machine that mixes living tissue, such as muscle, with man-made parts like a gel frame.

Muscle cells can contract, or squeeze tight, and that squeezing can push a fin, turning the cells' movement into motion for the robot.

Light works as a switch here: when light shines on the modified cells, they contract, so choosing where the light falls steers the robot.

The team built on an earlier muscle disk shaped by the iris, the coloured ring in the eye, learning how to grow thin muscle layers in patterns on a soft gel.

Break it down

LEAD: MIT scientists built a tiny swimming robot powered by living muscle instead of a motor
KEY EVENT: The gum-sized gel robot swims and steers using light-controlled muscle fins
+- WHO: MIT scientists, including study author Ritu Raman
+- HOW: Light makes muscle cells on a fin contract and flap, steering the robot and changing its speed
+- TEST: It swam and turned through a simple underwater maze
EVIDENCE: At its fastest it covered about four body lengths per minute, comparable to a cow shark swimming leisurely
BACKGROUND: The swimmer builds on an earlier iris-inspired muscle disk that moved only about 100 microns
OPEN QUESTION: Whether thin muscle robots can one day handle delicate jobs in fragile or unpredictable places

Why it matters

Thin, soft robots powered by living muscle could one day do delicate work in places too fragile or unpredictable for ordinary machines, and they may be cheaper and more efficient to build.

Test what you remember

1. What powers the robot's movement?

(A) A battery pack
(B) A bulky motor
(C) Living muscle cells
(D) A tiny propeller

2. About how big is the robot's frame?

(A) About the size of a shoebox
(B) About the length and width of a stick of gum
(C) About as long as a ruler
(D) About the width of a dinner plate

3. How do researchers steer the robot?

(A) By tilting the water tank
(B) By choosing which fin gets light and how often
(C) By pulling it with a string
(D) By shouting commands at it

4. How fast was the robot at its fastest?

(A) One body length per hour
(B) About four body lengths per minute
(C) 65 body lengths per minute
(D) 100 microns per minute

5. What did the earlier artificial muscle disk show?

(A) That gel frames could float on water
(B) That muscle cells could survive in space
(C) That thin muscle layers could be grown in patterns and controlled with light
(D) That a robot could swim across an ocean

6. Why does Raman think muscle-powered robots could be useful one day?

(A) Because they never need light
(B) Because living tissue is soft, responsive, and can heal itself
(C) Because they can swim faster than Olympic swimmers
(D) Because they are made of metal

Answer key (try the quiz first!): 1: C 2: B 3: B 4: B 5: C 6: B

Questions to think about

1. Study author Ritu Raman: Thinner muscle-powered designs could be cheaper to build, might move more efficiently, and could one day handle delicate jobs in fragile or unpredictable places.

MY THOUGHTS (at least 20 words):